

MOD-SEM SuSe25

Topic 3 - Phase 2

Hyperuniformity

Generation of hyperuniform structures and their analysis

Contents - Phase 2

- **Recap: Hyperuniformity**
- **Generating hyperuniform structures**
 - Generation of hyperuniform point patterns using the collective coordinate approach
 - Acceleration using nuFFT
 - Generation of hyperuniform gaussian scalar fields
- **Persistent Homology Analysis**
 - Analysis of modified a objective structure factor
- **Literature**

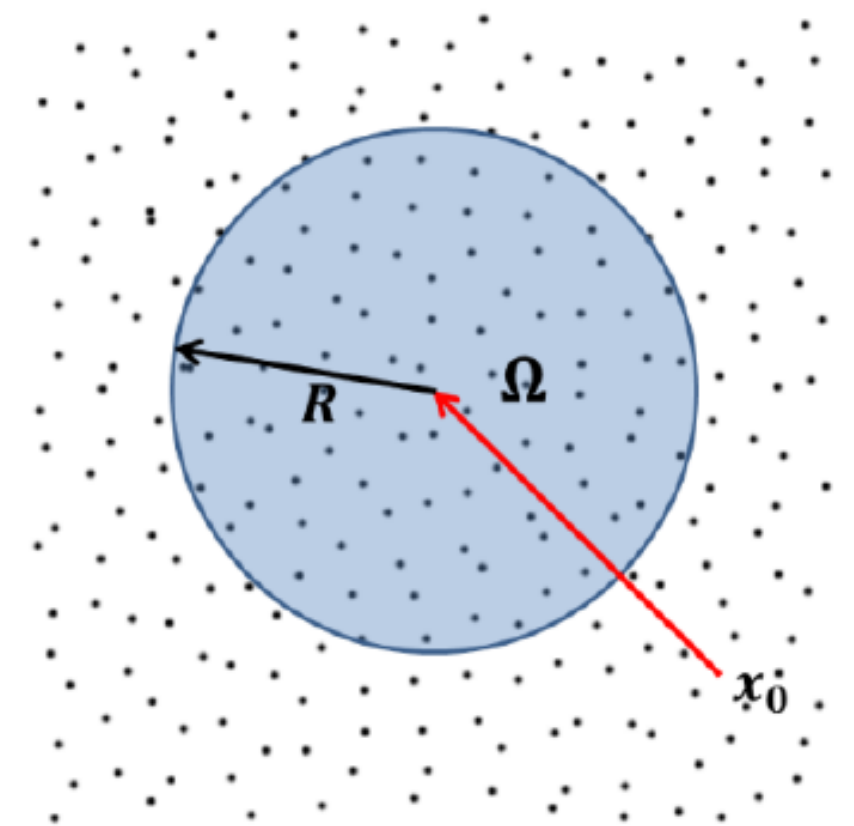
Recap: Hyperuniformity

- **Hyperuniform systems are defined by the suppression of density fluctuations over large spatial scales.**
- One can derive from internal correlation and fourier transform [Tor18] that hyperuniformity may be characterized by a vanishing structure factor for small wave vectors

$$\lim_{|k| \rightarrow 0} S(k) = 0$$

- For a single periodic configuration with a finite number of N points, the structure factor is identical to [Tor18]:

$$S(k) = \frac{1}{N} \left| \mathcal{F} \left(\sum_{j=1}^N \delta(r - r_j) \right) \right|^2 = \frac{1}{N} \left| \sum_{j=1}^N 1 \cdot \exp(ik \cdot r_j) \right|^2, k \in \mathcal{K}$$



Generating hyperuniform structures

Generation of hyperuniform point patterns using the collective coordinate approach

Generating hyperuniform structures

Generation of hyperuniform point patterns

- We may use a method based on optimization [SSDV24] towards a given structure factor

$$\min_r \mathcal{L}(r) = \min_{r_1, \dots, r_N} \sum_{k \in \mathcal{K}} |S(k) - S_0(k)|^2$$

- We can compute the derivative of the Loss analytically

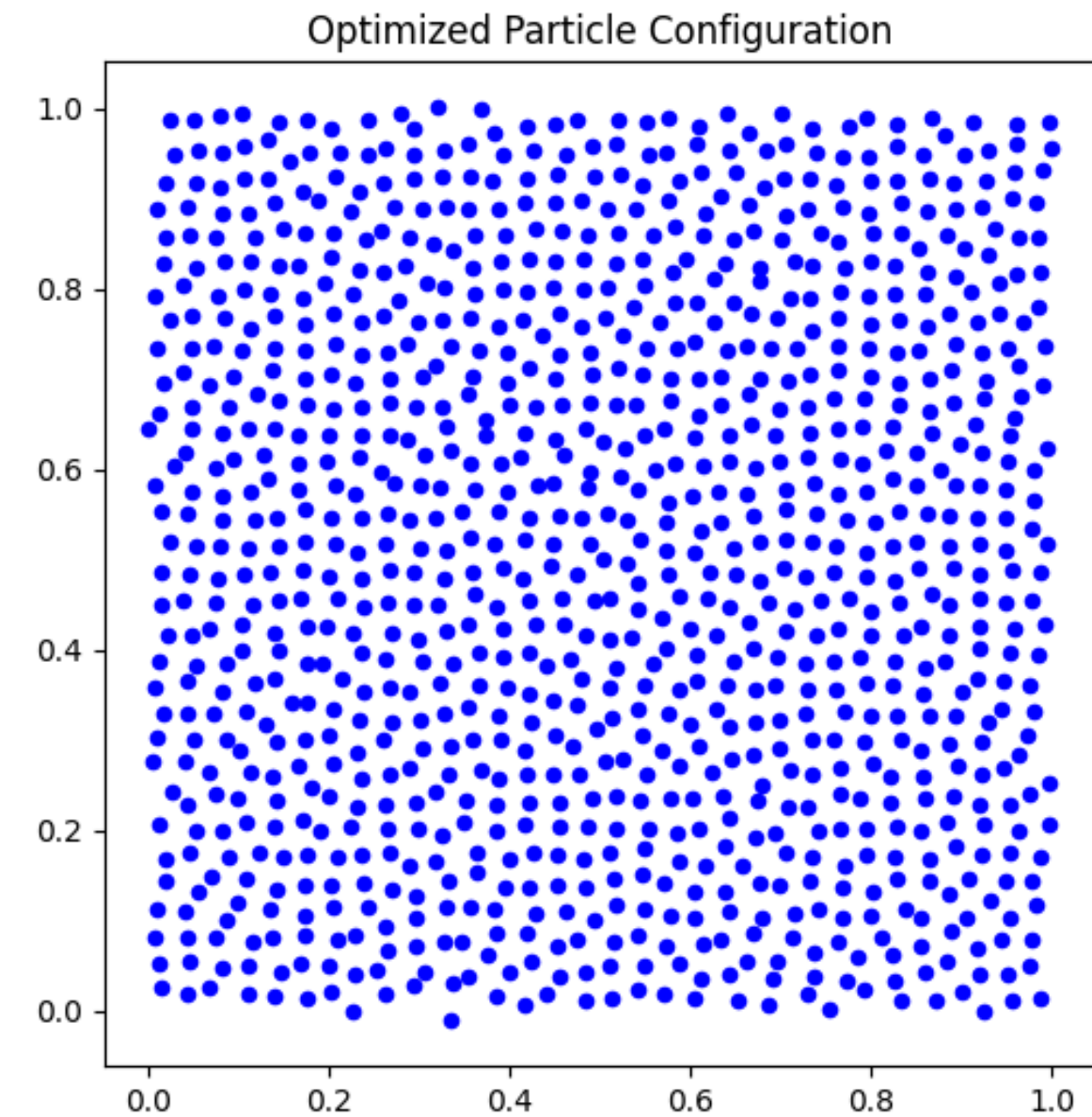
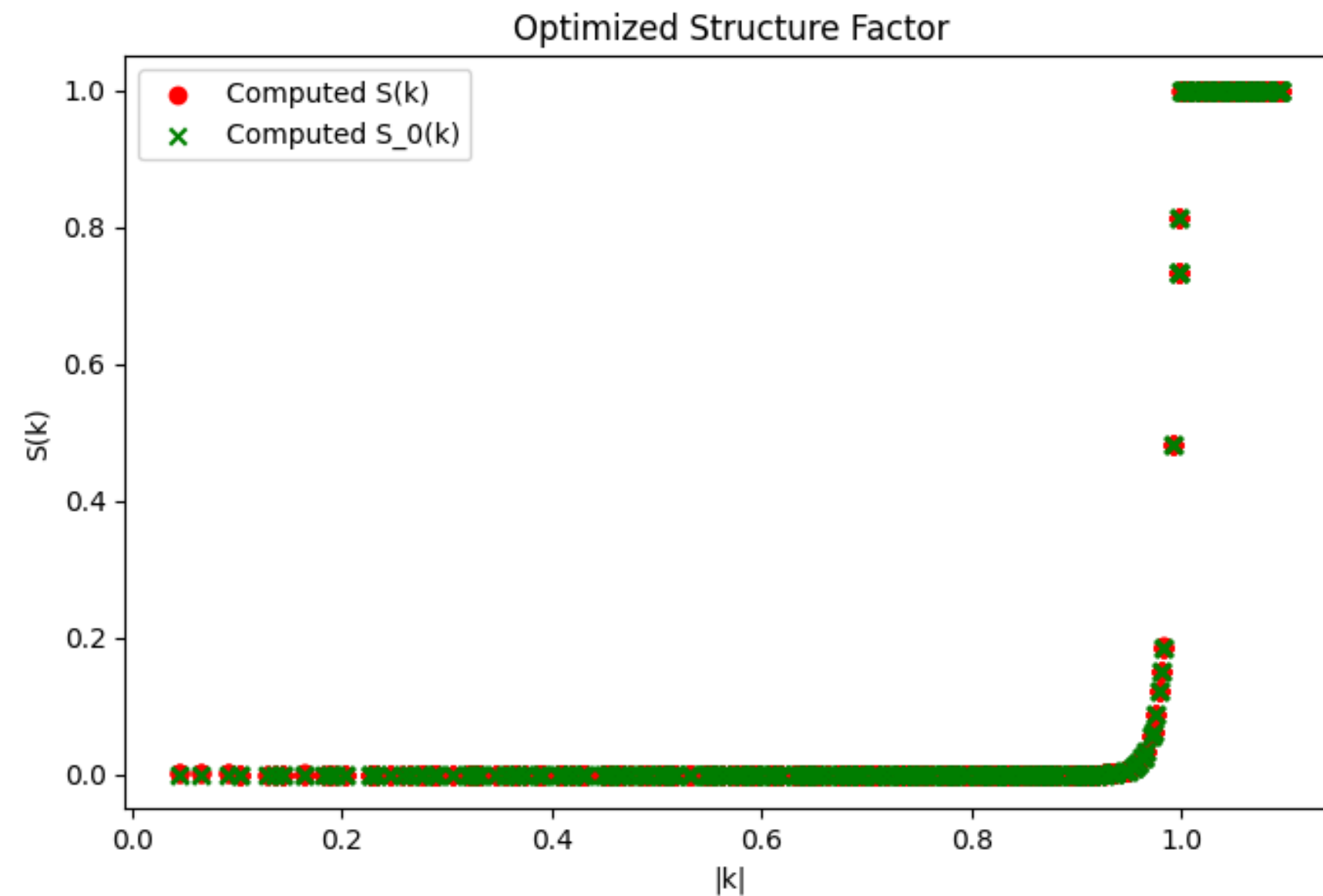
$$\frac{d\mathcal{L}}{dr_i} = \operatorname{Re}\left(\sum_{k \in \mathcal{K}} \frac{-4\mathbf{i}kw(k)}{N} (S(k, \mathbf{r}) - S_0(k)) \hat{\rho} \exp(-\mathbf{i}k \cdot r_i)\right)$$

- And feed it into a minimizer of our choice (e.g. SciPy's BFGS) together with an initial point configuration (e.g. randomly perturbed grid)

Generating hyperuniform structures

Generation of hyperuniform point patterns

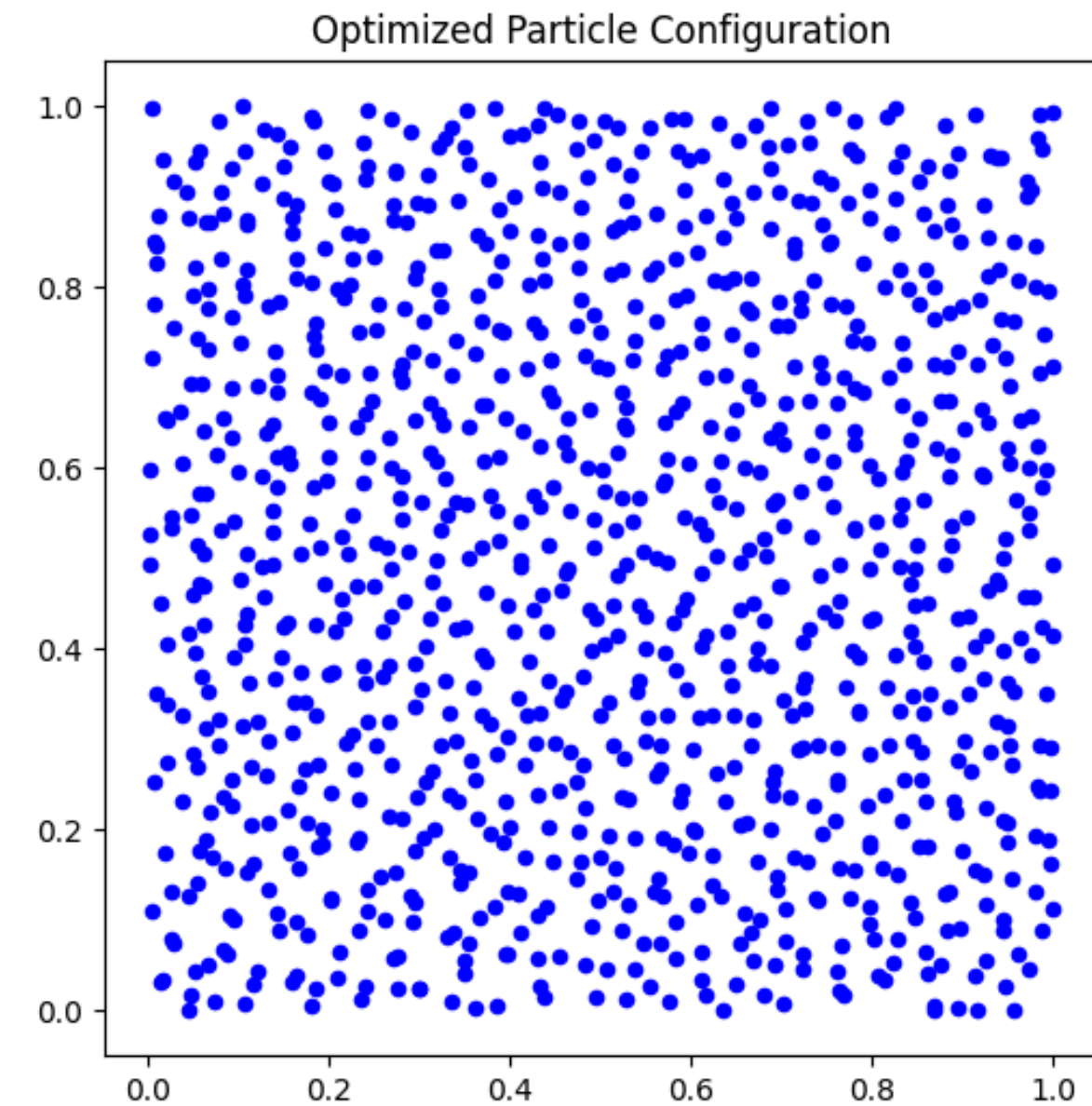
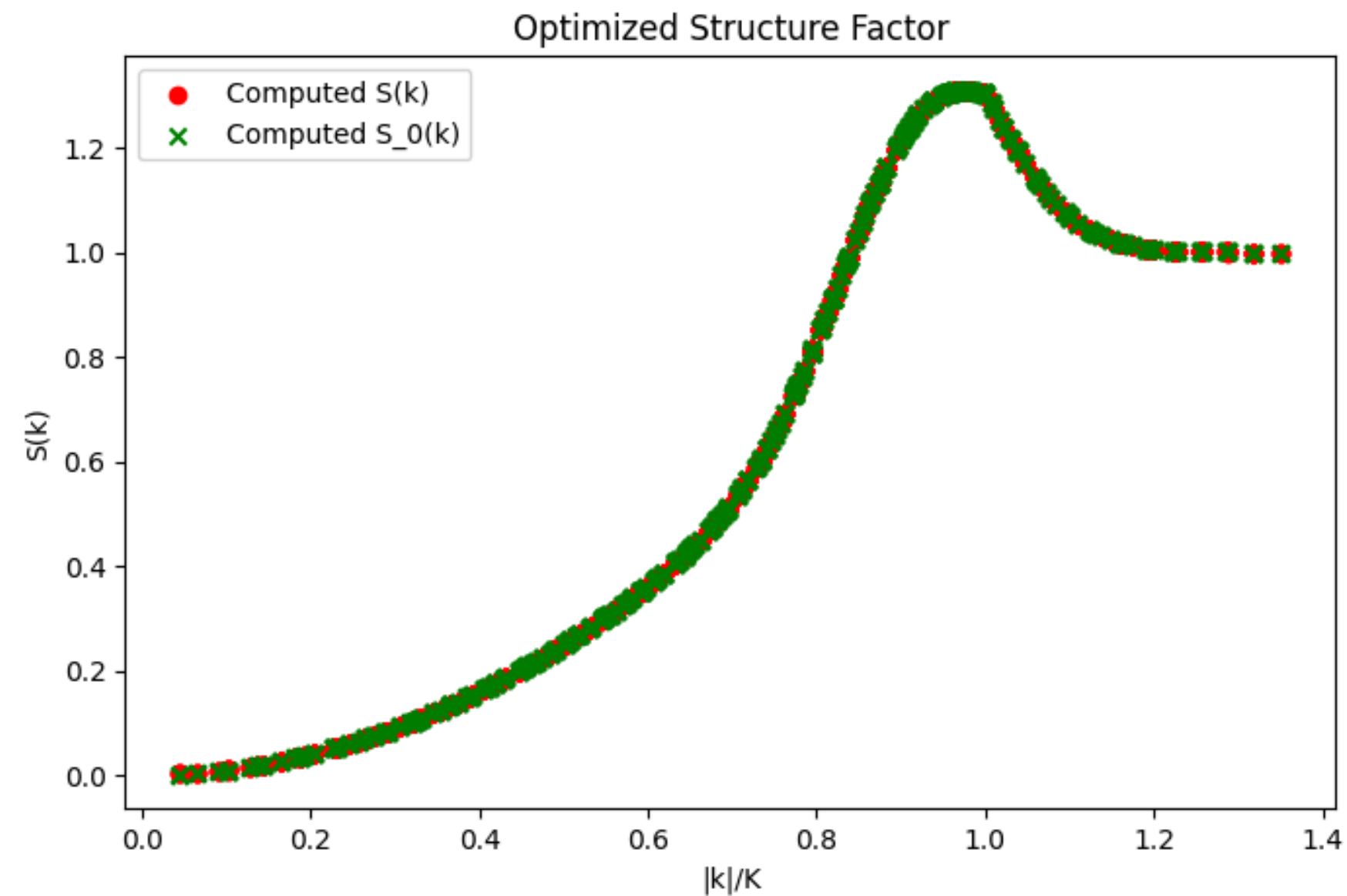
- We can easily generate different point patterns



Generating hyperuniform structures

Generation of hyperuniform point patterns

- We can easily generate different point patterns



Generating hyperuniform structures

Generation of hyperuniform point patterns - Acceleration using nuFFT

- In [SCM24] a scalable nuFFT-based approach to generating HU point patterns is proposed
- nuFFT is used to evaluate expressions of the form

$$f_j = \sum_{i=1}^I c_i \exp(\pm \mathbf{i} y_j \cdot x_i), \quad j = 1, \dots, J$$

- where $x_i \in \mathbb{R}^d, c_i \in \mathbb{C}, y_j \in \mathbb{R}^d$ in $\mathcal{O}(N \log N)$ [FIN24]
- Recall:

$$S(k, r) = \frac{1}{N} \left| \sum_{i=1}^N 1 \cdot \exp(\mathbf{i} k \cdot r_i) \right|^2$$

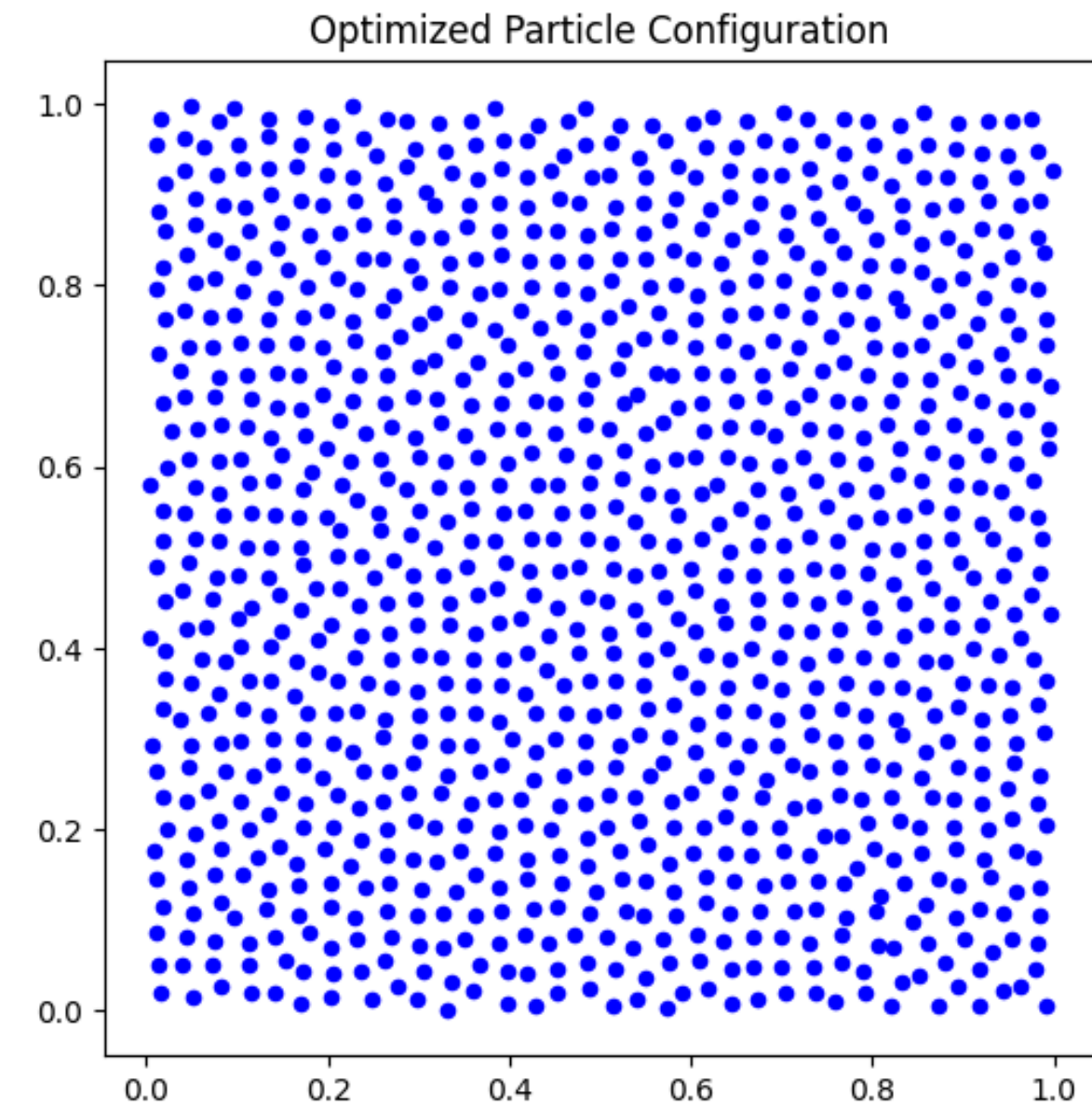
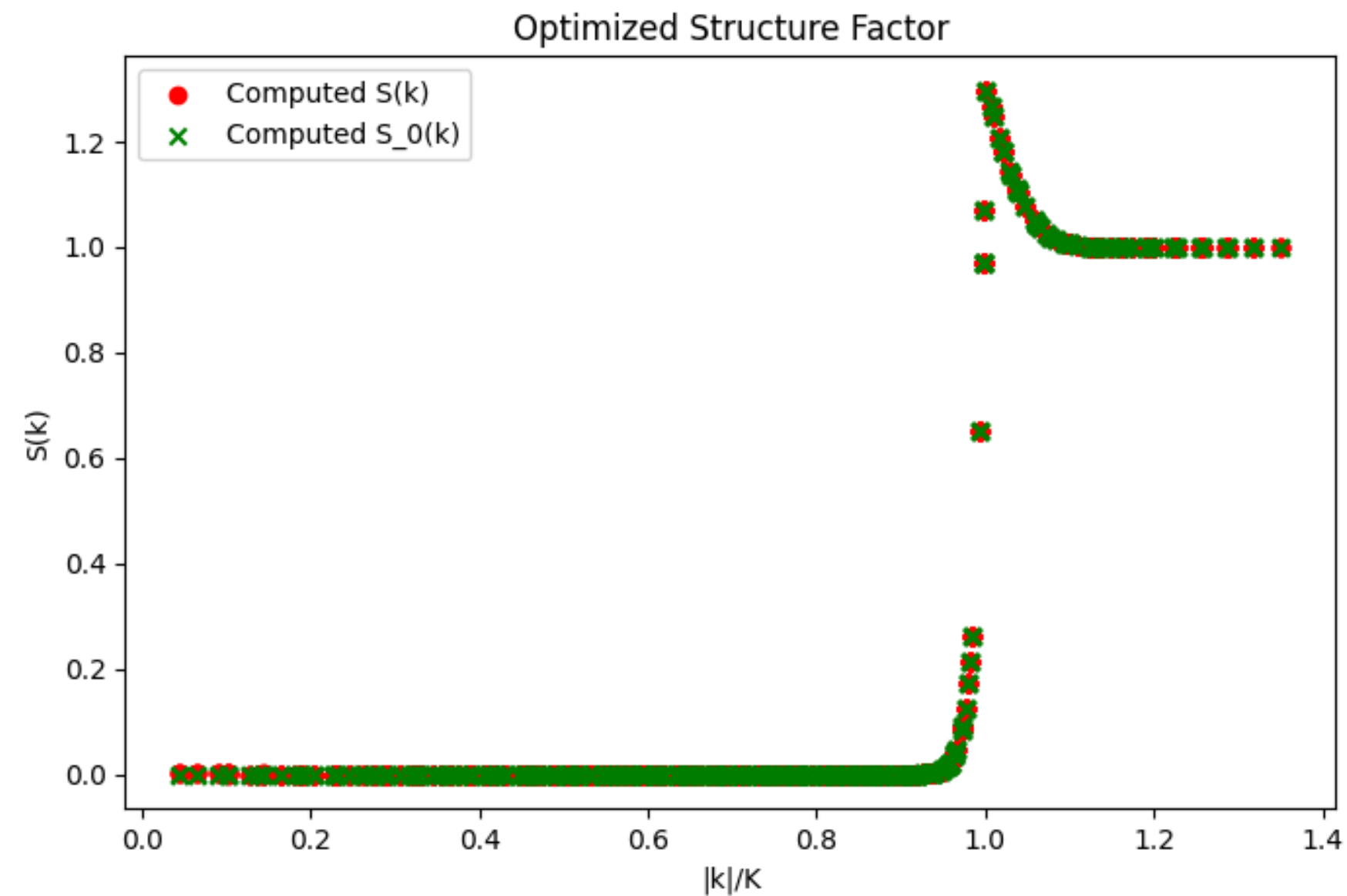
$$\frac{d\mathcal{L}}{dr_i} = \operatorname{Re} \left(\sum_{k \in \mathcal{K}} \frac{-4\mathbf{i} k w(k)}{N} (S(k, \mathbf{r}) - S_0(k)) \hat{\rho} \exp(-\mathbf{i} k \cdot r_i) \right)$$

- Therefore we can compute the structure factor and the derivative of the loss function using nuFFT

Generating hyperuniform structures

Generation of hyperuniform point patterns - Acceleration using nuFFT

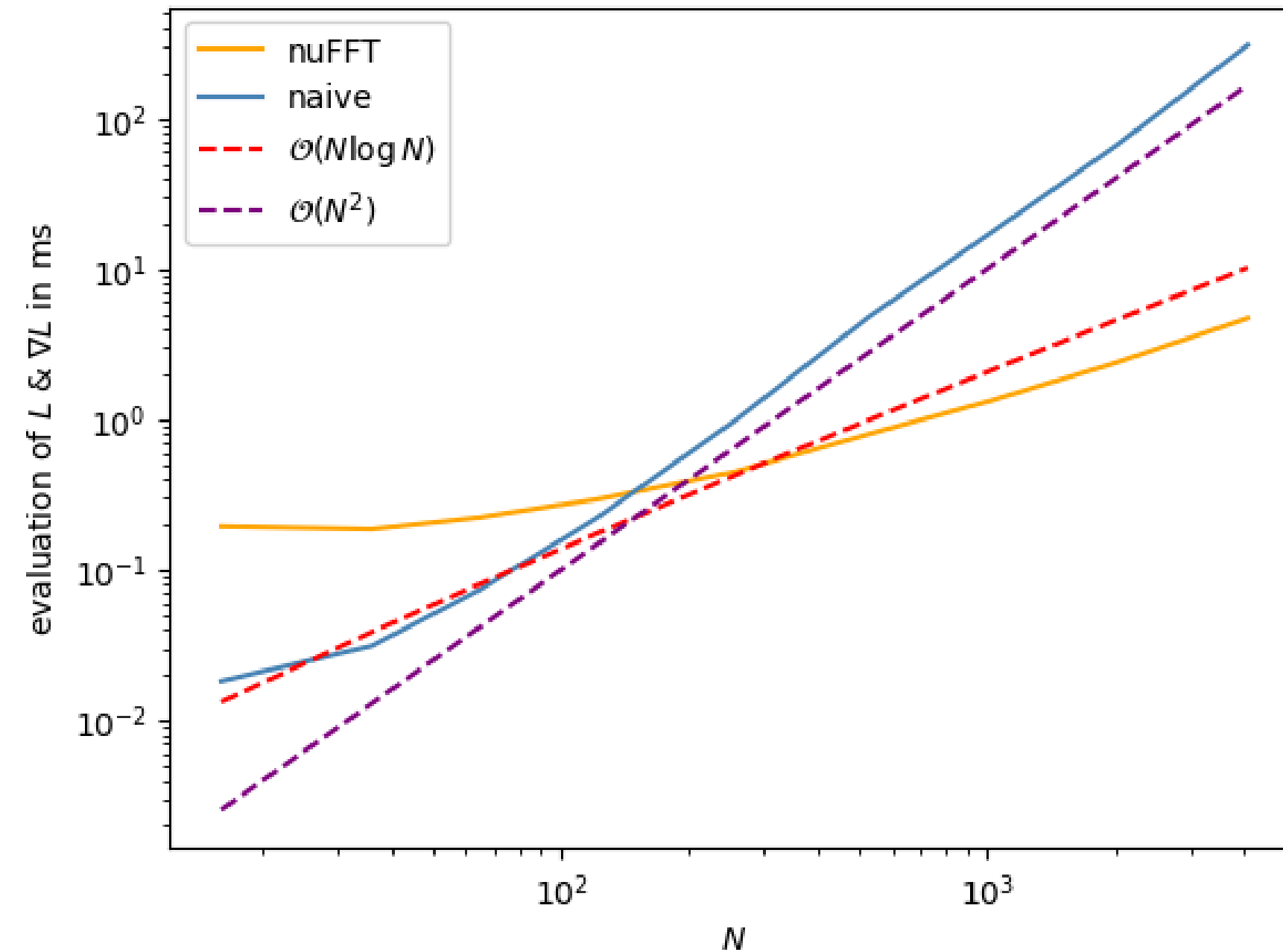
- We can once again generate point patterns



Generating hyperuniform structures

Acceleration using nuFFT - Benchmark

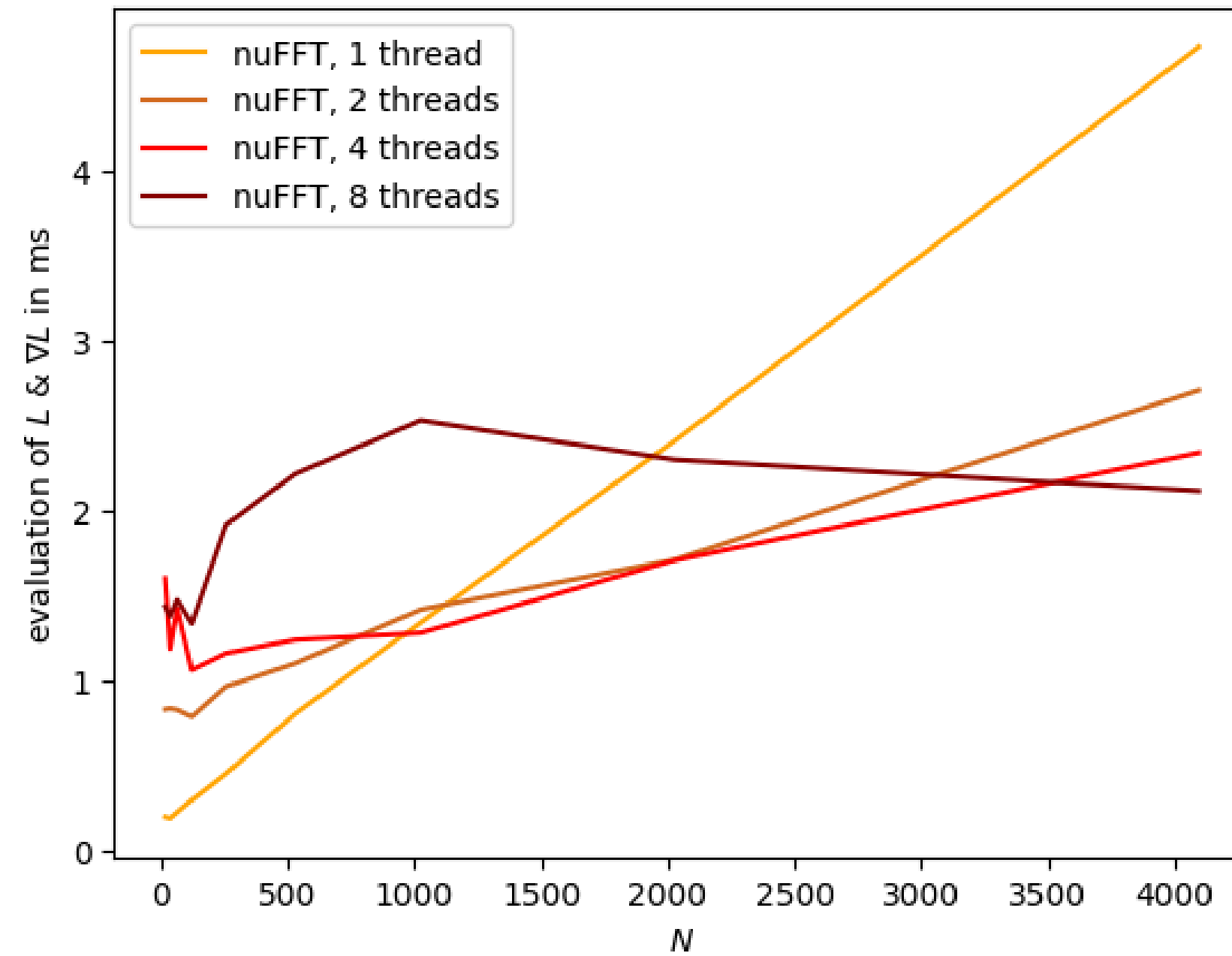
- We benchmarked the two methods



Generating hyperuniform structures

Benchmark - Multithreading

- We benchmarked the method with multithreading



Generating hyperuniform structures

Benchmark - Full Simulation

- We observe faster generation times of patterns of $N = 1024$ points

	naive	nuFFT
Function evaluations	1519	1004
Gradient evaluations	1519	1004
Iterations	1000	1000
Loss value	$\approx 2.6 \cdot 10^{-4}$	$\approx 3.0 \cdot 10^{-4}$
Time	11m 24.2s	6m 47.6s

Generating hyperuniform structures

Generation of hyperuniform gaussian scalar fields

Generating hyperuniform structures

HU scalar fields

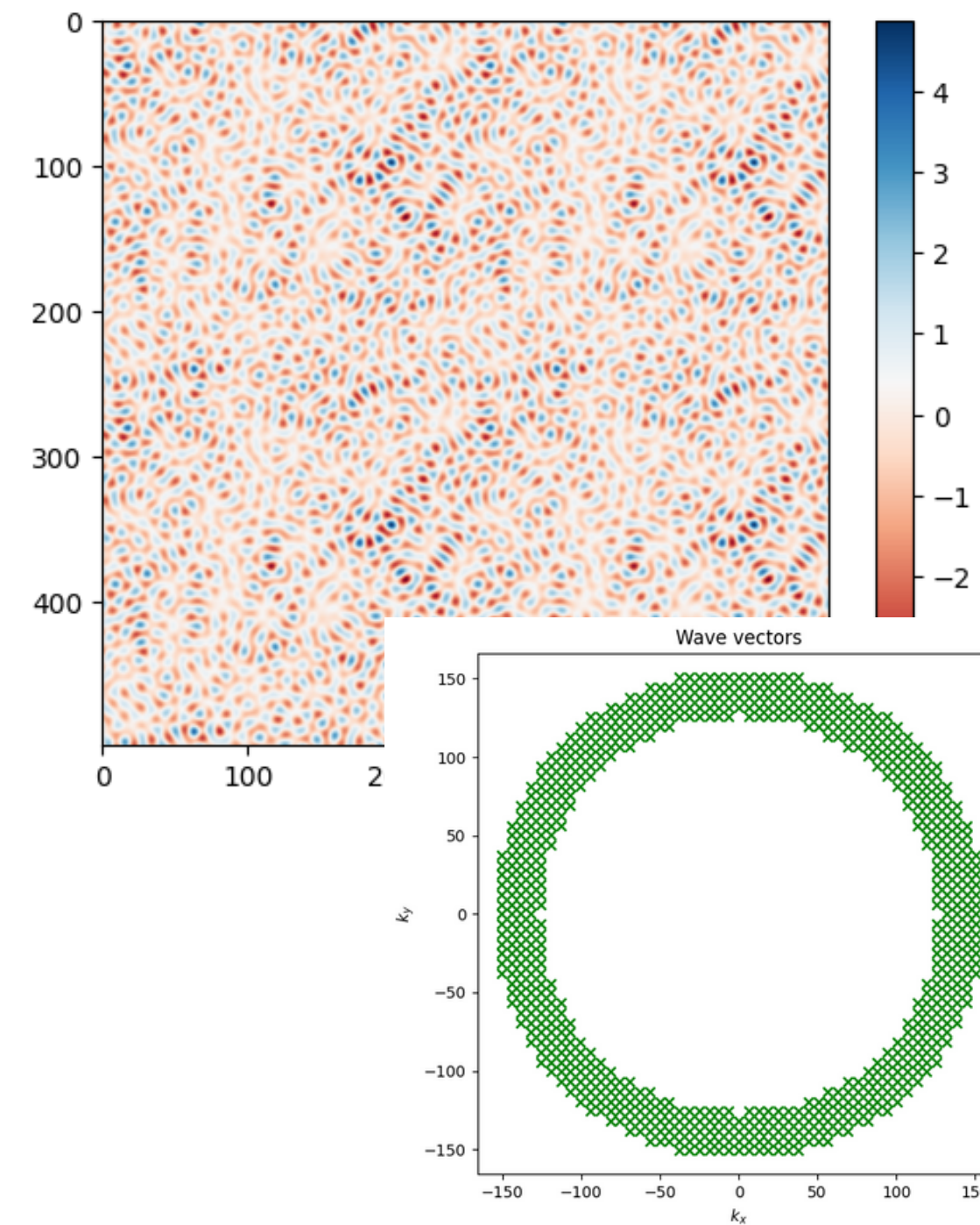
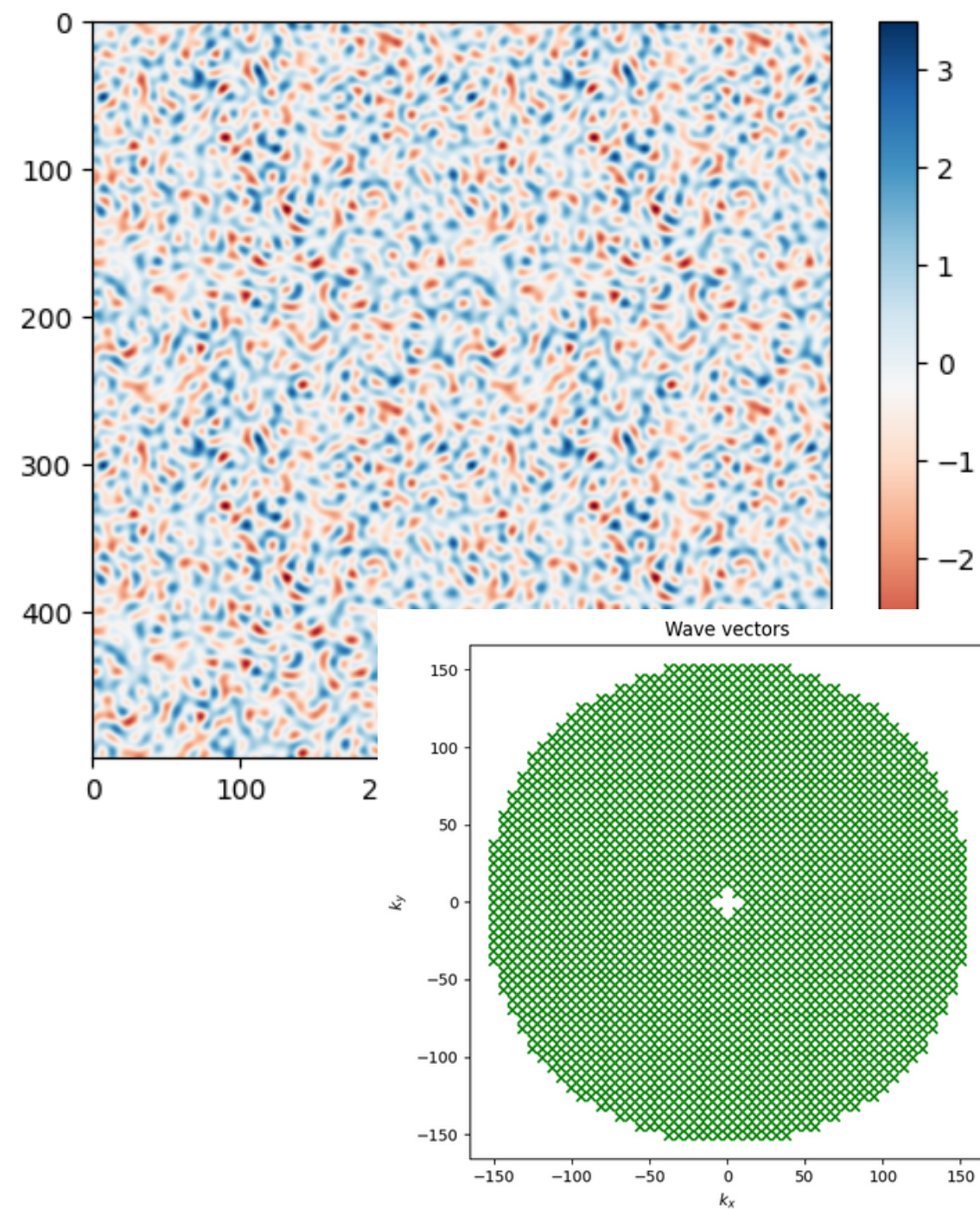
- Hyperuniformity can be extended to scalar fields
- We may be able to extend analysis of HU point patterns to analysis of HU scalar fields

Task: Generate hyperuniform scalar field from a gaussian random field [MT17], Section III & IV

- Current progress:
 - Prototype: Able to generate gaussian random fields with a certain spectrum
 - We still use constant amplitude instead of an dynamic amplitude based on the wave vector k

Generating hyperuniform structures

HU scalar fields



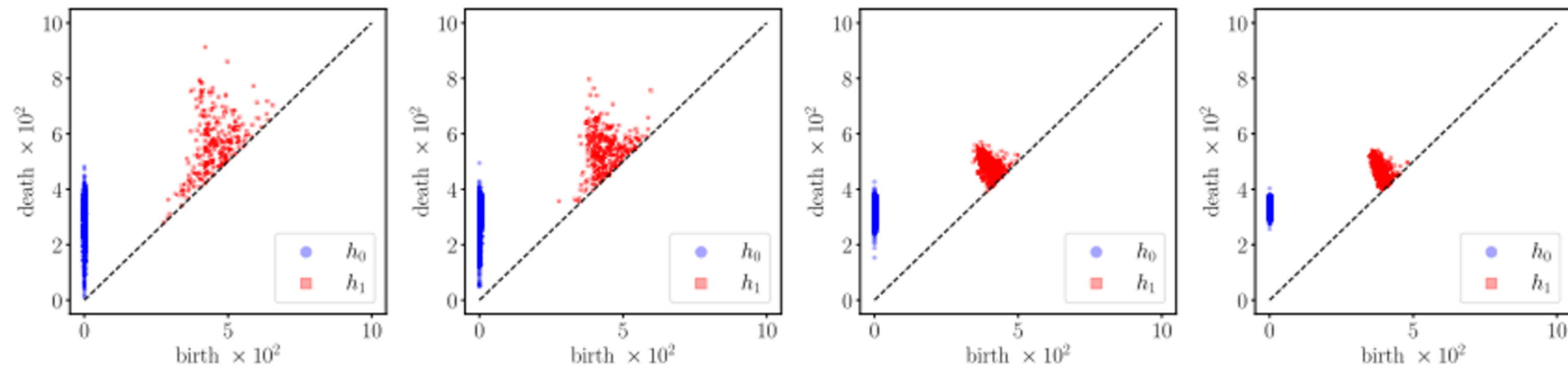
Persistent Homology Analysis

Persistent Homology Analysis

Persistent Homology and Topological Analysis

- A Vietoris–Rips complex is built by growing balls of radius r around each point
- Homological features are computed: h_0 (connected components), h_1 (loops)
- Persistence diagrams and histograms of death times are analyzed to capture topological features.

Task: Use *ripser* or *GUDHI* to compute persistent homology and analyze the resulting diagrams



Persistent Homology Analysis

Topological Comparison and Robustness to Finite Sampling

- Wasserstein distances $W1, W2$ are used to compare persistence diagrams:
 1. Across different pattern parameters
 2. Between full systems and cropped subsets

$$W_p(A, B) = \inf_{\gamma} \left(\sum_{\mathbf{x} \in A} (\|\mathbf{x} - \gamma(\mathbf{x})\|_q)^p \right)^{1/p},$$

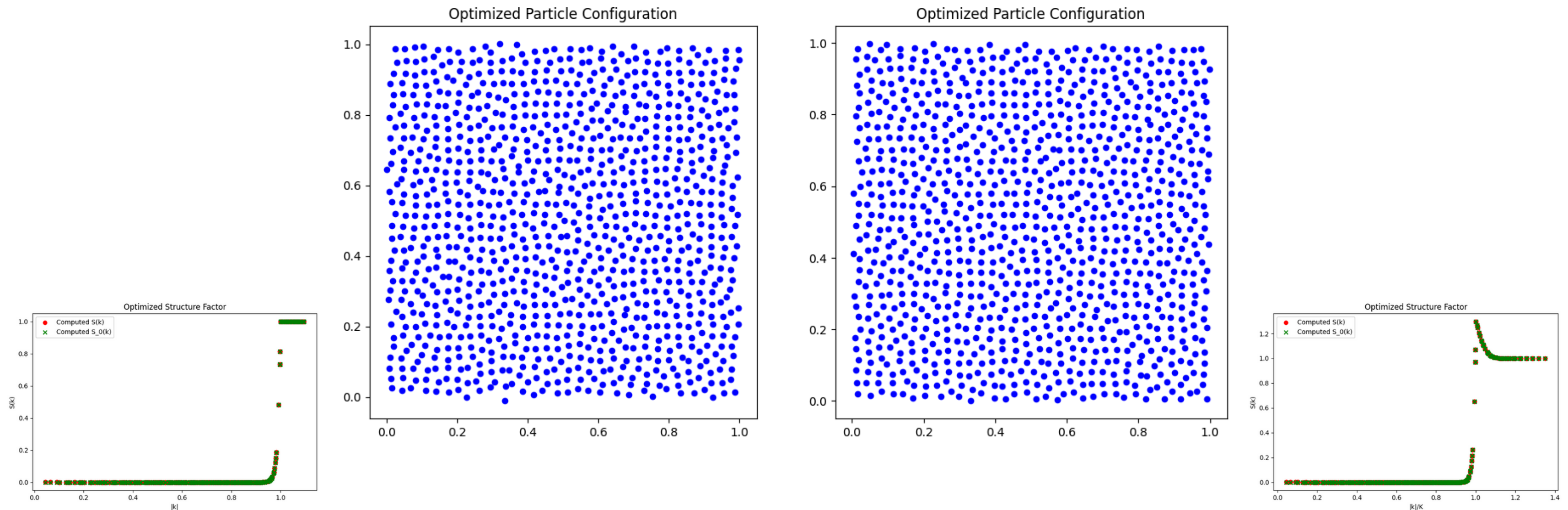
- The study shows that HU properties are preserved in subsets above a certain size

Task: Compare diagrams using Wasserstein distance, identify HU regions in parameter space, and analyze robustness to partial data

Persistent Homology Analysis

Analysis of modified a objective structure factor

- We may slightly change the objective structure factor with a bump (as indeed is often observed in real-world experiments [Tor18])
- We have already seen two point configurations with the only difference being a bump - visually no discernable difference



Persistent Homology Analysis

Analysis of modified a objective structure factor

- Can we quantify the difference using the persistent homology?
- We compute for random initial perturbations the mean and variance within the no bump group to eachother and between the bump and no bump group

	no bump vs. no bump	no bump vs. bump
h_0	$[0.192; 1.1 \cdot 10^{-3}]$	$[0.264; 3.3 \cdot 10^{-3}]$
h_1	$[0.231; 6.0 \cdot 10^{-4}]$	$[0.258; 6.3 \cdot 10^{-4}]$

- Conclusion: The difference is small but discernible, especially in the h_0 component.
- Illusiveness of hyperuniformity can effectively be analyzed using the persistent homology!

Literature

Literature

- [Tor18] Salvatore Torquato. Hyperuniform states of matter. *Physics Reports*, 745:1–95, 2018. Hyperuniform States of Matter.
- [SSDV24] Marco Salvalaglio, Dominic J. Skinner, Jörn Dunkel, and Axel Voigt. Persistent homology and topological statistics of hyperuniform point clouds. *Phys. Rev. Res.*, 6:023107, May 2024.
- [MT17] Zheng Ma and Salvatore Torquato. Random scalar fields and hyperuniformity. *Journal of Applied Physics*, 121(24):244904, 2017.
- [SCM24] Aaron Shih, Mathias Casiulis, and Stefano Martiniani. Fast generation of spectrally shaped disorder. *Phys. Rev. E*, 110:034122, Sep 2024.
- [FIN24] FINUFFT Contributors. FINUFFT Mathematical Formulation. Accessed: 2025-06-04. Flatiron Institute Nonuniform Fast Fourier Transform (FINUFFT). 2024. url: <https://finufft.readthedocs.io/en/latest/math.html>.